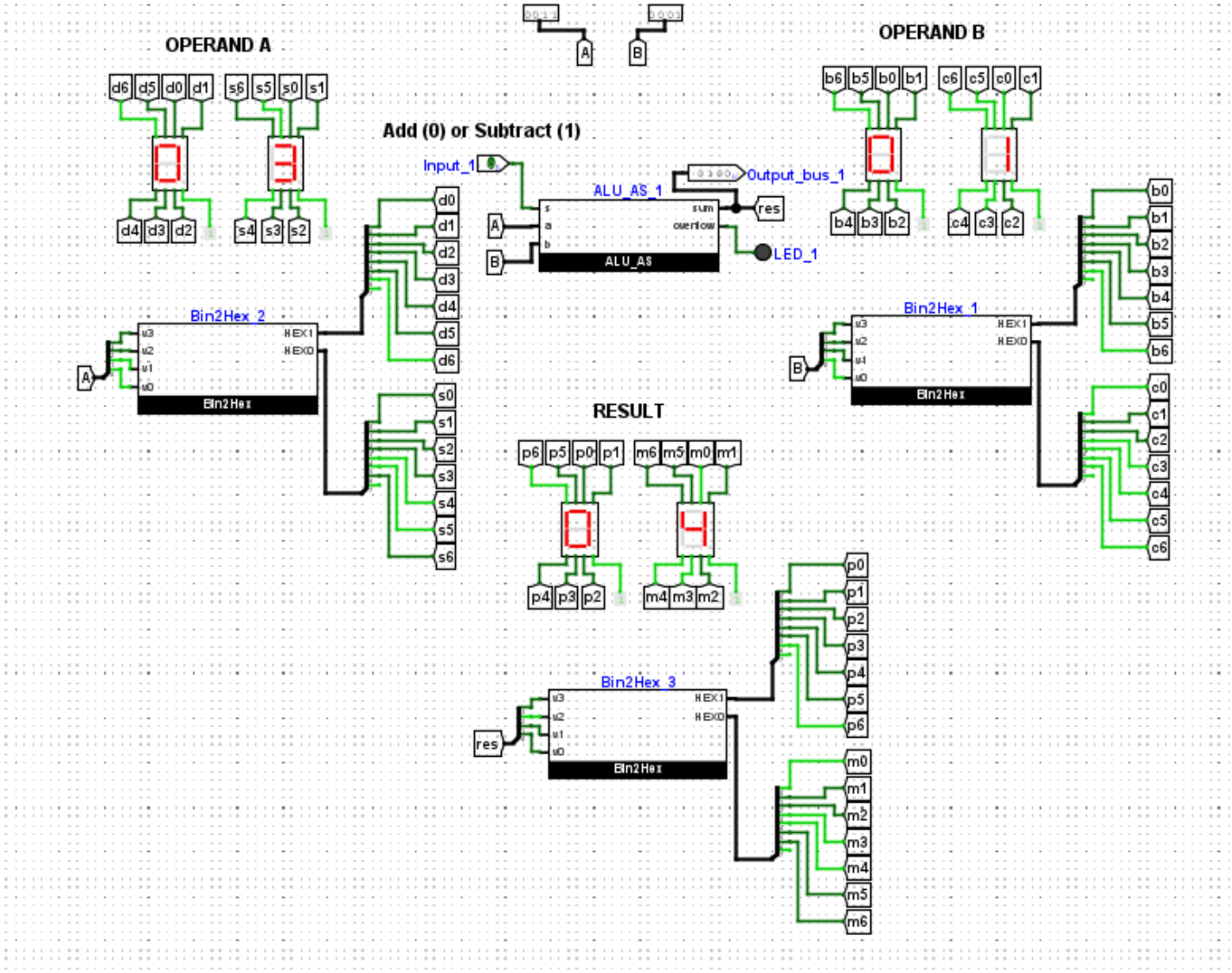
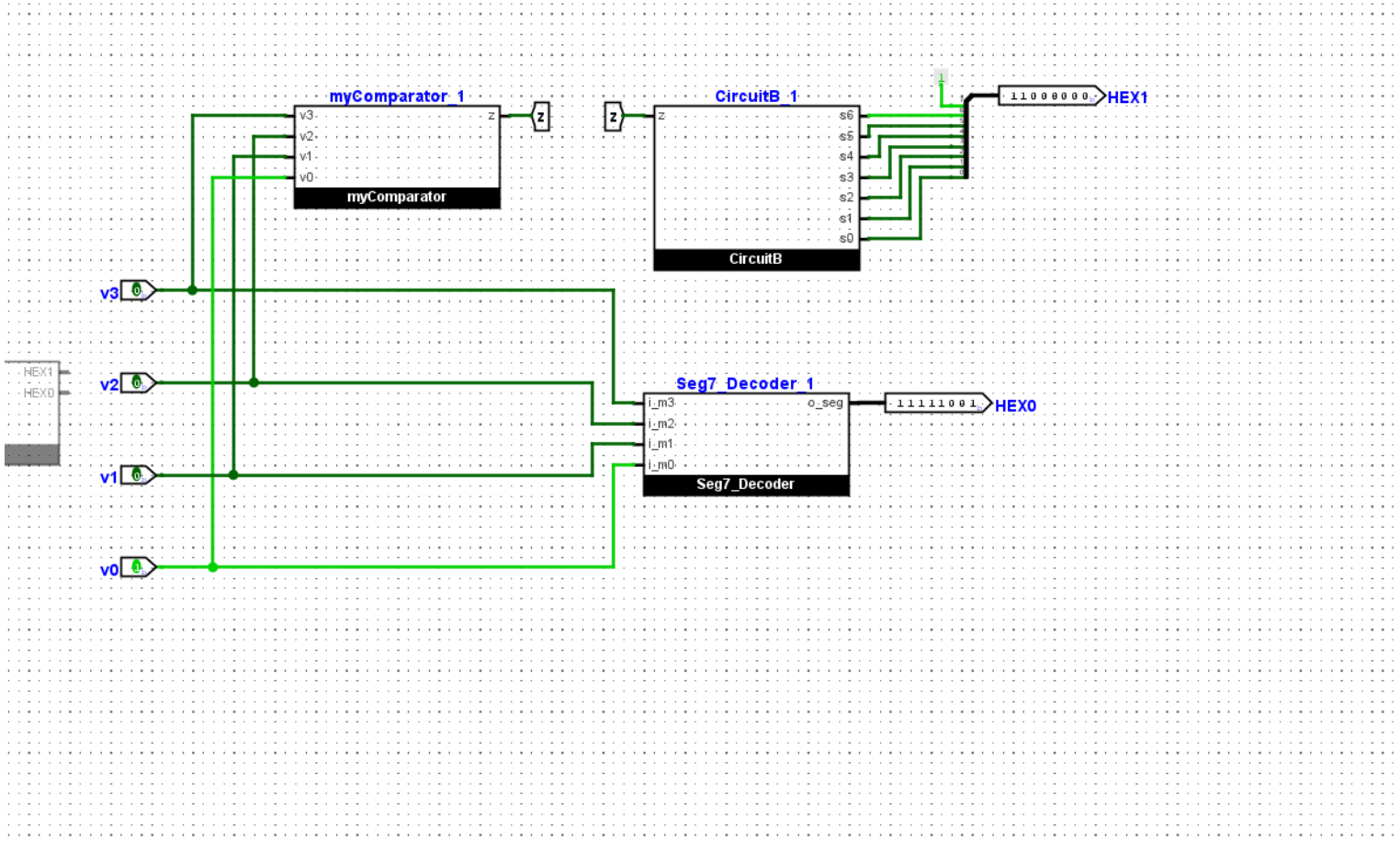
14 bit calculator

By Nicklaus Brown

14 bit calculator circuts

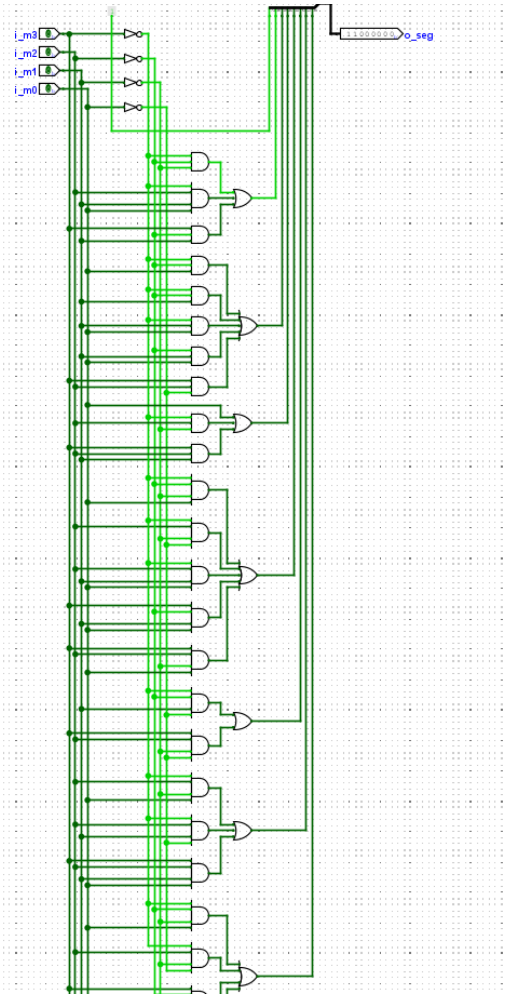


The 14 bit calculator is built off these three main components which are reused for the final calculation



Seg_7 Decoder

- Translate the 4 bit digits into the seven segment display
- Which is controlled by seven different control signals from 1 – 7.
- When 4 bit matches to a 1, then the one control signal is activated to on. (Check second to last slide for visual representation)

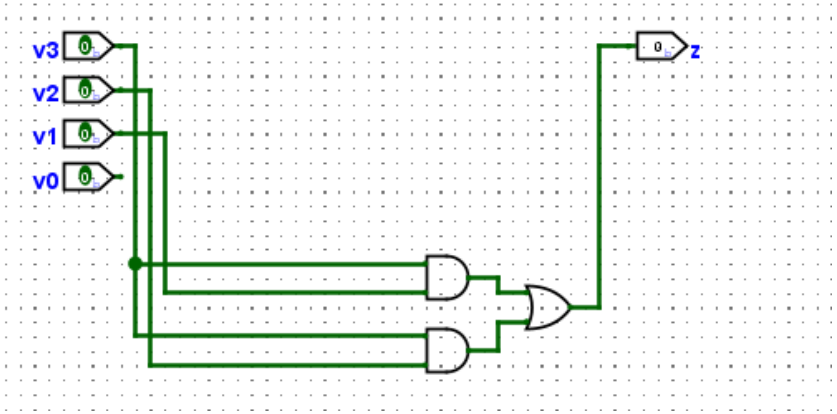


i_m3	i_m2	i_m1	i_m0	o_seg[7..0]
0	0	0	0	1 1 0 0 0 0 0 0
0	0	0	1	1 1 1 1 1 0 0 1
0	0	1	0	1 0 1 0 0 1 0 0
0	0	1	1	1 0 1 1 0 0 0 0
0	1	0	0	1 0 0 1 1 0 0 1
0	1	0	1	1 0 0 1 0 0 1 0
0	1	1	0	1 0 0 0 0 0 1 0
0	1	1	1	1 1 1 1 1 0 0 0
1	0	0	0	1 0 0 0 0 0 0 0
1	0	0	1	1 0 0 1 0 0 0 0
1	0	1	0	1 1 0 0 0 0 0 0
1	0	1	1	1 1 1 1 1 0 0 1
1	1	0	0	1 0 1 0 0 1 0 0
1	1	0	1	1 0 0 1 1 0 0 1
1	1	1	0	1 0 1 1 0 0 0 0
1	1	1	1	1 0 0 1 0 0 1 0

The comparator

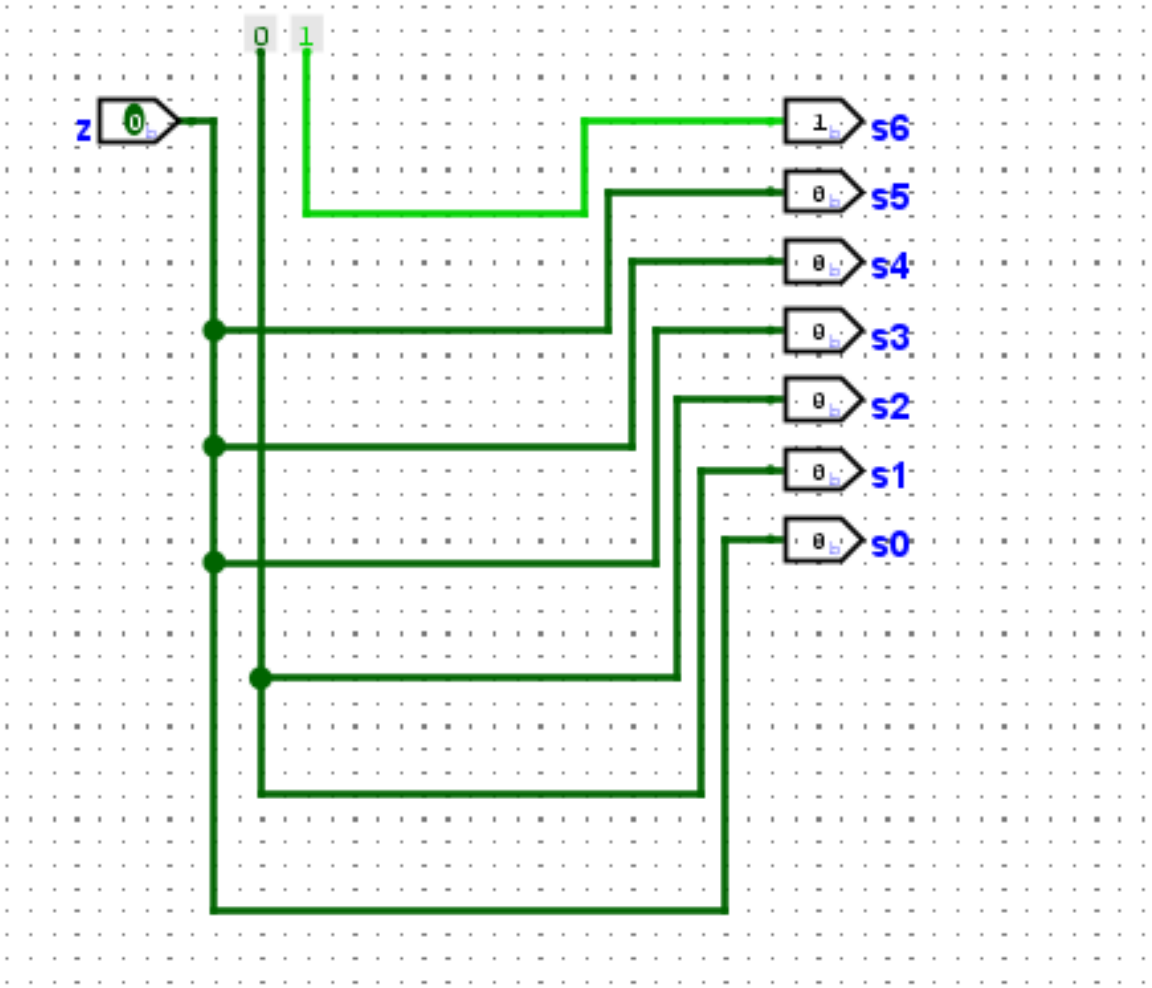
- The comparator compares the 4 bit binary against the value 9
- If the 4 bit binary is bigger then 9, then the output is 1 true. But if any else then it returns 0 false.
- Serving as the control signal for circuit B and drives the ten digits on the 7-segment display.

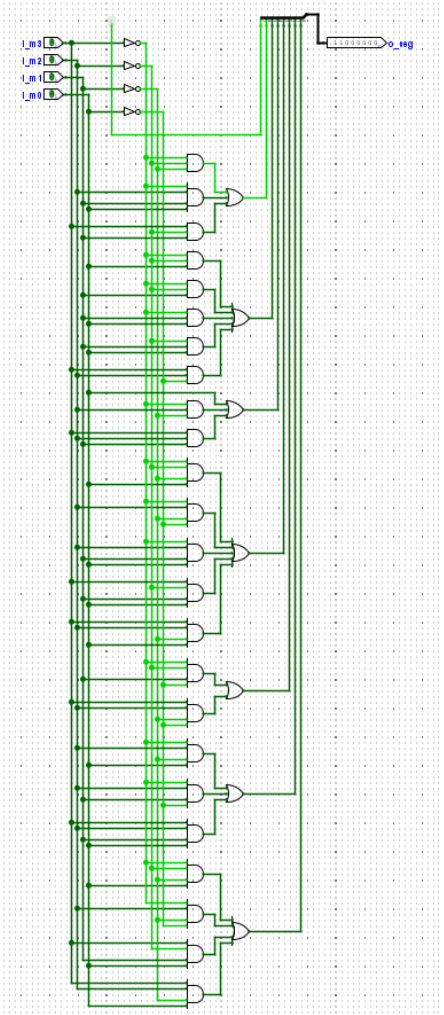
v3	v2	v1	v0	z
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

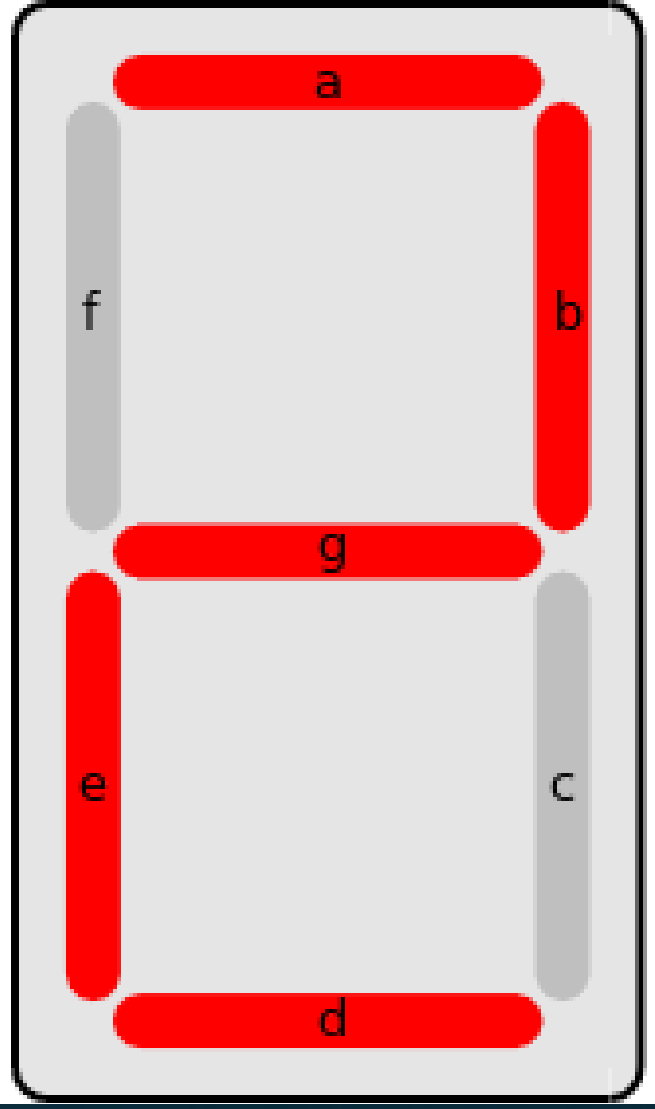


Circuit B

- Circuit B acts as the converter on any number that exceeds 9, which means it turn any letter past 9 back into a number to be added or subtracted.
- It does this by either adding 6 or subtracting 10 to drive the units digits display







16 of 16 rows shown

3	i_m2	i_m1	i_m0	o_seg[7..0]
	0	0	0	1 1 0 0 0 0 0 0
	0	0	1	1 1 1 1 1 0 0 1
	0	1	0	1 0 1 0 0 1 0 0
	0	1	1	1 0 1 1 0 0 0 0
	1	0	0	1 0 0 1 1 0 0 1
	1	0	1	1 0 0 1 0 0 1 0
	1	1	0	1 0 0 0 0 0 1 0
	1	1	1	1 1 1 1 1 0 0 0
	0	0	0	1 0 0 0 0 0 0 0
	0	0	1	1 0 0 1 0 0 0 0
	0	1	0	1 1 0 0 0 0 0 0
	0	1	1	1 1 1 1 1 0 0 1
	1	0	0	1 0 1 0 0 1 0 0
	1	0	1	1 0 0 1 1 0 0 1
	1	1	0	1 0 1 1 0 0 0 0
	1	1	1	1 0 0 1 0 0 1 0

The seven segment display which controls the clock as in 1 = off and 0 = on (replace all letters with number from 1 to 7.

The Calculation

- Both the ALU_AS/RC_4bits are responsible for taking two-4- bit binary inputs and does the computation using XOR gates and active carry ins to give the binary result of A + B.

- The 4- bit sum is then flowed through the Comparator and circuit B to split the value into two separate decimal units. Which is then displayed on a 7 segment decoders

